

The Synergy between Big Data and the Internet of Things

The Internet of Things generates fast streams of useful data. The challenge before enterprises is to store the vast amounts of data and to make the best use of it. This is where Big Data plays an important role.



An increasing number of gadgets now use smart technologies to generate data through embedded sensors. A car with smart apps installed in it, a smart home device that monitors the temperature indoors, a fitness tracker that sends the steps of a workout routine to your phone's app—all these are examples of the Internet of Things (IoT). These devices are connected to the Internet. It is estimated that by 2020, there will be 24 billion IoT devices across the world, which would naturally result in the generation of massive volumes of data. The digital universe is set to reach 40 zetta bytes by 2020. So, IoT delivers the information while Big Data acts on it to derive insights that will render these devices the precursors of a new technological age.

What is IoT?

IoT (Internet of Things) is a network of interconnected devices such as computers, cars, smartphones, kitchen

appliances, heart monitors, etc. As technology advances, even gadgets with the most basic functions like a watch, heart pacemakers, remote controls, etc., will have embedded sensors capable of collecting and exchanging data over the Internet. These can be controlled by a remote device. The sensors and chips generally gather data but don't process it. They send it to another place for analysis. Data on the performance of smart gadgets and customer usage patterns is generally gathered and analysed.

Components of IoT

IoT ecosystem: The IoT ecosystem includes all the elements such as a dashboard, remote, gateways, the network, security and storage, which allow devices to be connected to their users—businesses, government and consumers.

Entropy: These are the users such as businesses, government and consumers, who use the devices and generate